

# EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol

## An Earth observation that survives an agent handoff.

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence. Where no observation exists, emem signs an absence and its reason, not a zero.

A compact citation keeps the observation traceable across agents and time. Design goal: encode in orbit, decode in AI's reasoning with emem.

Jaya Kumari · Avijeet Singh · Vortex AI · avijeet@vortex.ai · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)  
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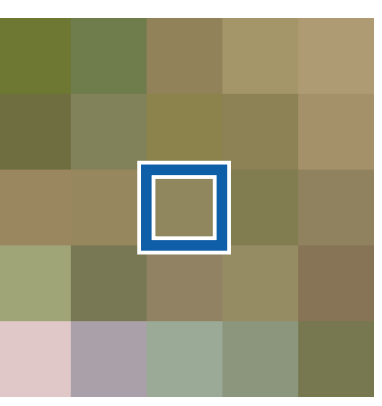
### From observation to the next agent

| Observe                                       | Locate                                       | Record                                      | Hand off                               | Resolve                                             | Check                                                              | Continue                                                  |
|-----------------------------------------------|----------------------------------------------|---------------------------------------------|----------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|
| Satellite or sensor<br>Keep the source scene. | Location · product · time<br>Lookup identity | Canonical record + CID<br>Batch attestation | Pass the reference<br>MCP · REST · A2A | Retrieve the same record<br>Recover its provenance. | Hash · bind · sign · source<br><b>Policy: refuse if one fails.</b> | Reason with cited evidence<br>Carry the reference onward. |

1 Does agent B act on a corrupted handoff?

## The same corruptions, five handoff conditions.

Agent A cites a satellite reading. A relay or a faulty signer changes the evidence. Agent B acts on it, declines it or uses the genuine record.

| Agent A cites                                                                                                                                                                                                                             | handoff                                                                                        | relay                                                                                                                                                                                                                                       | Agent B                                                                                                                                                                                       | B did not act on corrupted evidence agents, pooled Claude | declined   | used the genuine record |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------|-------------------------|
| <br><b>Agent A reads NDVI 0.4709 and cites it</b><br>Sentinel-2A L2A, 25 Sep 2026<br>outlined: the 10 m pixel the record names<br>RGB from recorded grids | <b>prose</b><br>The Sentinel-2 L2A NDVI for the 10 m cell at Keylong, Laha...                  | <div>relay or faulty signer<br/>one change per trial,<br/>shared corruption families:</div> <ul style="list-style-type: none"><li>value · unit · cell · time · band · source · derivation · pixel · stale record · signature / id</li></ul> | <b>B</b> reads the text                                                                                                                                                                       | <b>22 / 276</b>                                           | <b>22</b>  | <b>0</b>                |
|                                                                                                                                                                                                                                           | <b>JSON</b><br>{ "about": { "field": "Keylong, Lahaul (32.57126 N, 77.03448...                 |                                                                                                                                                                                                                                             | <b>B</b> reads the fields                                                                                                                                                                     | <b>112 / 276</b>                                          | <b>106</b> | <b>6</b>                |
|                                                                                                                                                                                                                                           | <b>retrieved text (RAG)</b><br>Field log, Keylong, Lahaul (32.57126 N, 77.03448 E), cell ...   |                                                                                                                                                                                                                                             | <b>B</b> retrieves a passage                                                                                                                                                                  | <b>56 / 276</b>                                           | <b>56</b>  | <b>0</b>                |
|                                                                                                                                                                                                                                           | <b>opaque id</b><br>record ref obs-7d1c44                                                      |                                                                                                                                                                                                                                             | <b>B</b> fetches the sender's record                                                                                                                                                          | <b>134 / 288</b>                                          | <b>88</b>  | <b>46</b>               |
|                                                                                                                                                                                                                                           | <b>emem reference + verifier</b><br>emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk6... |                                                                                                                                                                                                                                             | <div><b>B</b><ul style="list-style-type: none"><li>resolve</li><li>re-hash</li><li>place / time / band</li><li>signature</li><li>log</li><li>recompute</li><li>source re-read</li></ul></div> | <b>300 / 300</b>                                          | <b>264</b> | <b>36</b>               |

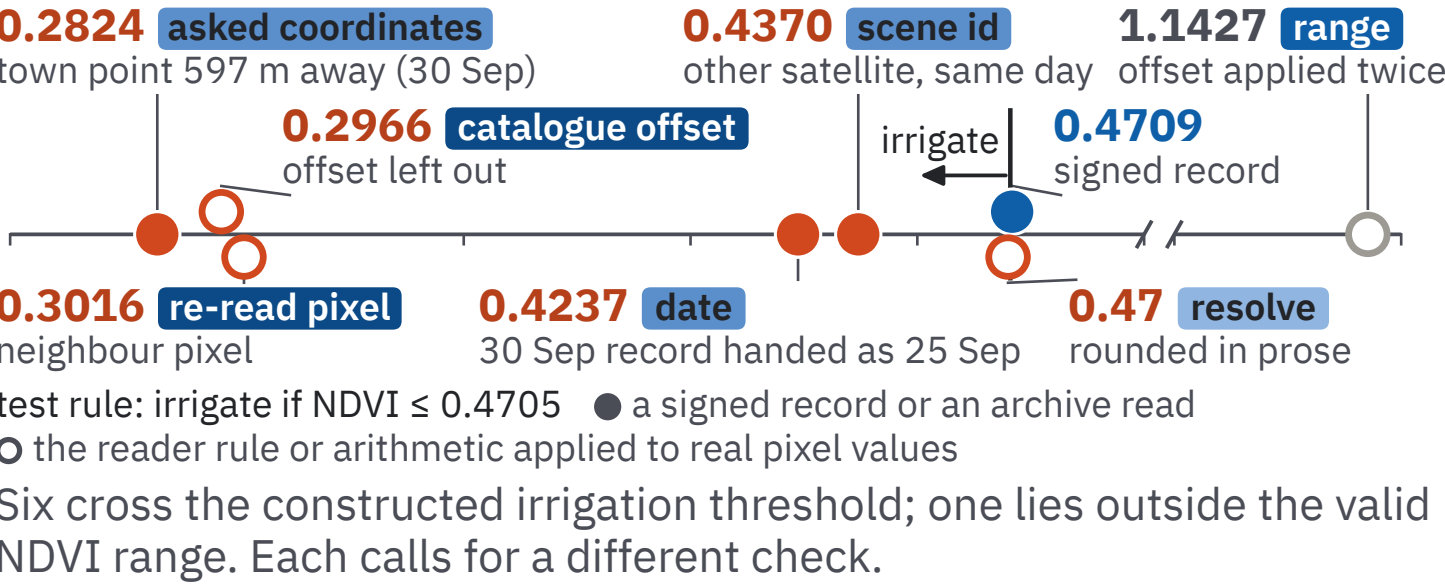
Across the 23 shared items, agents with an emem reference did not act on 276 of 276 corrupted handoffs (240 declined, 36 used the genuine record); with an opaque id, 122 of 276; in prose, 22 of 276.

By design, Earth data stays on the encoding device in orbit; its signed record and reference are downlinked, so no source re-read (SAT-042 harness, not flown).

2 How does a value change between agents?

## One NDVI, eight values

We varied one Keylong NDVI by scene, date, place, pixel and arithmetic.



3 Why can a value change?

## A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

### CONCEPTUAL CHANGE MODEL

$$\Delta Z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

$\Delta z$ : change in readout. The terms name possible causes (environment, sensor, geolocation, encoder, residual); we have not measured their shares.

7 What does a checked reference establish?

## The record, not physical truth

| CHECKABLE                       | NOT ESTABLISHED           |
|---------------------------------|---------------------------|
| Record bytes + content identity | Sensor accuracy           |
| Declared cell, band, time       | Entity identity           |
| Attestation signature           | Physical truth            |
| Declared derivation             | Decision correctness      |
| Recorded history / as-of        | Source quality: inherited |

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

### FULL TRACE · 30 SEP 2026

**Token** **record** **source pixel**

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG[row, col] = DN(b) & Ed25519(α) & leaf in STH & STH<sub>geo</sub> q<sub>a</sub> prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked 15 links (17 checks) in 17.9 s, all verified. Grey: what must still be trusted after each check.

**1-3 content address** 1,115 B hash to the fact\_cid; one flipped mantissa bit gives mu5x2cks... · still trusted: BLAKE3 and the verifier's code

**4-5 cell binding** hashed cell = token cell = the lat/long the derivation used · the cell rule

**6-7 derivation** scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem's code

**8-9 upstream** the STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · this STAC item carries no checksum

**10-11 server signature** emem.dev's Ed25519 receipt over the request id, time, cells and returned fact\_cids · shows what emem.dev served, not that the value is right

**12-14 public log** entry 2,457,078 is in the 2,556,451-entry log, consistent with the signed tree head (STH) geo.qa co-signed · geo.qa is also run by Vortex AI

**15 signing key** the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

Links 1-5 and 12-14 fit in a 4,906-byte file that reads 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

### THE CONTRIBUTION

#### Checkable observation handoff

EO evidence that travels between agents, checked by the receiver down to the source pixel.

Content hashes (IPFS), signed logs (SCITT) and provenance (PROV) already exist. emem adds typed EO references and a source re-read, which caught a wrong pixel that hashing, signatures, logs and recomputation had passed (panel 5).

## 12 Use emem. Carry the evidence forward.

|                                                                   |                                                                                             |                                                                                                                    |                                                                                                            |                                                                                                                             |                                                                                                         |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>ChatGPT · @emem plugin</b><br>"Signed facts about real places" | <b>Claude · Directory plugin / MCP</b><br>"hand another agent a token it can check offline" | <b>Diffy · Marketplace plugin</b><br>"Shared memory for AI agents. One address per fact, one signature you check." | <b>GitHub MCP Registry · Server listing</b><br>"Satellites for AI."                                        | <b>ClawHub · OpenClaw plugin listing</b><br>"Shared, signed and content-addressed memory of the physical world..."          | <b>CONNECT</b><br> |
| Enable emem; paste a token.                                       | Resolve from chat or terminal.                                                              | Workflow templates, including EUDR                                                                                 | Install MCP server                                                                                         | Install the OpenClaw plugin.                                                                                                | Setup · EO tasks<br>Reproduce                                                                           |
| <b>MORE CLIENTS</b><br>Cursor · VS Code · Gemini CLI              | <b>PROTOCOLS</b><br>MCP · A2A<br>REST / OpenAPI                                             | <b>DEVELOPER</b><br>Python · TypeScript · Docker<br>LangGraph memory · GitHub source                               | <b>FRAMEWORK EXAMPLES</b><br>LlamaChain · LlamaIndex · AutoGen<br>CrewAI · Mastra · Agno (see setup notes) | <b>DISCOVERY / MIRRORS</b><br>MCP Registry · MuleSoft Exchange · Glama · Smithery<br>Context7 · Hugging Face (older mirror) |                                                                                                         |